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“Rural exploitation” in solar energy development? A field survey experiment in South Korea on solar energy support in rural areas

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ABSTRACT

As renewable energy facilities are predominantly installed in rural areas, the geographical mismatch between the production and consumption of renewable energy is prominent. In various cases of rural opposition to renewable energy development, rural opponents often translate this mismatch into the narratives of “rural exploitation” to mobilize further opposition. These narratives highlight that renewable energy development places growing burdens on rural communities, thereby provoking a sense of distributive injustice among rural residents. Using a regression analysis with data collected from a field survey experiment in South Korea, this study shows that the narratives of rural exploitation decrease support for solar energy among rural residents, particularly if they live in more remote, economically underprivileged rural areas. Our finding highlights that a broader historical context of rural-urban inequality should guide our understanding of the motivations and conditions of rural opposition to renewable energy development.

1. Introduction

Many studies have investigated various sources of local opposition to renewable energy (RE) development [1,2]. The underlying idea is that RE facilities, mainly utility-scale solar and wind energy facilities, provide diffused benefits (of low-carbon energy production) but impose concentrated costs on local communities [3–5]. Such costs include negative impacts on the local environment [6], landscape [7–11], and health (particularly noise nuisances from wind turbines) [12–14]. Given that most utility-scale RE facilities concentrate in *rural* areas [15], some studies have focused on a broader context of rural-urban inequality in understanding the sources of local opposition [16–18]. Nilson and Stedman [18] point out that rural opposition is partly shaped by the perceived ‘rural burden’ imposed by urban needs, leading to the sense of distributive and procedural injustice associated with RE development.

A handful of cases of rural opposition illustrate this perception of rural burden. One resident who opposes the Lund Hill Solar Farm proposed in Klickitat County, WA, noted, “[t]his community is going to bear the brunt of all the consequences – there are many – but none of the benefits except taxes” [19]. Another resident in Copake, NY, in response to the New York State’s Accelerated Renewable Energy Growth and Community Benefit Act (AREGCB) passed in April 2020, stated, “[t]he way the state has structured this law, the towns who will bear the burden

of these very large installations are not given any say” [20]. Wind energy opponents in the Great Lakes region [21,22] view that “big developers are getting rich off of the project on the backs of local communities” [23].

Examples reach beyond the US. Ko [24] interviewed a resident in a South Korean rural county who said, “[w]hile most of the electricity flows to cities, solar farms are in forests and farmlands in rural areas, and they have no benefits for us” (p. 4). Delicado et al. [25] conducted interviews with 150 residents in 4 rural municipalities in Portugal hosting RE projects, finding that negative opinions largely stem from perceived distributive injustice. One of the interviewees noted, “[i]t is a pity that the electricity costs don’t go down here... that we don’t have any benefit from the wind farm” (p. 80). Lennon and Scott [26] documented a narrative used by a wind opposition group in Westmeath, Ireland, stating: “Westmeath has become the focal point for the entire country in opposition to what is clearly a scheme designed to profit a small number of people while damaging the majority of people’s [sic] right to peaceful enjoyment of their home and environment” (p. 98).

These narratives of rural burden may not represent rural residents’ general views on RE development. Only a few well-organized groups that oppose RE development in rural areas may subscribe to and (re) produce the notion of rural burden [27,28]. Some may argue that these narratives misrepresent reality and spread misinformation, as often

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condemned by RE developers [29]. However, these narratives may reflect a broader pattern of ‘rural exploitation’ that rural residents perceive, not only in energy [30] but also in economic and industrial development [107–110]. That is, rural residents may perceive that RE development ‘exploits’ rural communities that shoulder the burdens of hosting RE projects with local negative externalities, whereas urban communities can gain access to the clean energy they produce without shouldering such burdens. Viewed in this way, the narratives of rural burden can decrease rural residents’ support for RE development by provoking a sense of distributive injustice between rural and urban areas.

This study examines whether the narratives of rural exploitation in solar energy development affect rural residents’ support for solar energy. It employed a field survey experiment in South Korea where scholars point to the role of ‘exploiting rural’ in bringing about industrial economic growth, particularly because of the export-oriented industrial policy that has squeezed rural areas [31,32]. South Korea is also a valuable area of research not only because it is understudied in the literature on RE acceptance but also due to its rural-urban inequality in electricity production-consumption [27]. We experimented in two areas with different levels of ruralness to examine whether residents respond differently to the narratives.

While numerous studies have examined the role of narratives in shaping RE acceptance [33–36], few have explored what narratives should undermine it at a local level, especially in rural areas where researchers report growing cases of local RE opposition. Building upon earlier research that emphasizes the role of rural-urban inequality in understanding rural opposition [18], this study contributes to the literature by examining whether the notion of rural exploitation in RE development undermines rural support for it. It also aims to explain why opposition groups in rural areas continue using the narratives of rural burden in RE development, thereby enhancing our understanding of their opposition tactics and the motivations behind them.

2. Literature review

2.1. Rural exploitation in renewable energy development

Utility-scale RE facilities, both solar and wind, are predominantly proposed and installed in rural areas with affordable land prices because of their land-intensive nature [37–41]. This implies that local opposition to RE development is often ‘ruralized’ [42,43], i.e., shaped by the historical context of rural-urban spatial dynamics [44,45]. Even though the sources of rural opposition to RE development are highly location-specific, the socioeconomic relationship between rural and urban areas helps understand those sources.

Scholars of critical social acceptance research and energy justice literature have been keen on these dynamics [46–52]. Both strands of research have long argued that local opposition to RE development should not be reduced to “NIMBYism,” criticizing that it fails to capture various structural underpinnings, such as socioeconomic and institutional contexts, that shape local acceptance of RE development [3,53]. Batel [54] points out that the recent literature on RE acceptance paid close attention to the mode of neoliberal capitalism in RE development and how it shapes people’s responses to RE technology. Viewed in this way, local opposition to RE development is not to simply ‘free-ride’ on diffused public goods with concentrated costs [3], but to contest a broader socio-political and economic system that entrenches exploitative practices against local communities in the name of RE development and climate mitigation.

Rural-urban spatial dynamics is a key example of these structural underpinnings. According to rural studies, in the initial stage of industrial development, urban and rural areas begin to undertake different economic specializations based on their factor endowment. That is, urban areas with abundant labor resources specialize in industrial production, and rural areas with ample land and natural resources

specialize in agricultural production. Scholars in this literature argue this makes rural areas less prosperous than urban areas, either due to lower productivity in the agricultural sector compared to non-agricultural (i.e., industrial) [55] or through ‘natural resource dependence’ [110]. Development studies scholars argue further that governments with export-led industrial policy can structurally impose this rural poverty by squeezing the rural agricultural sector, such as by forcing low grain prices, to keep labor costs lower for global competitiveness [107–109] (but see Varshney [56]). Even in the later stage of industrial development, studies show that rural areas remain underprivileged due to the negative and divisive impacts of the tourism-dependent rural economy [57], yet with mixed results [58,59]. These arguments lead to the concept of ‘rural exploitation,’ where rural economies are (re) configured primarily to serve urban needs, creating unequal economic development between rural and urban areas.

Rural exploitation is not limited to the economic domain. Kelly-Reif and Wing [60] examined rural environmental injustice [61] in two cases where rural communities experienced environmental degradation from energy production (in the Fukushima Daiichi nuclear accident case) and from industrial agriculture (in the Eastern North Carolina case) that urban populations do not. They further note that polluting industries in these cases ‘bought out’ rural communities to accept environmental burdens in exchange for public goods such as infrastructure, health, and education, to which they generally lack access compared to urban populations. Aragón and Rud [62] also found evidence in Ghana where large-scale gold mining released environmental pollutants that negatively affected the total factor productivity of agricultural activities in the neighboring rural communities.

Though relatively understudied, RE injustice may mirror rural exploitation. Because RE facilities are also accountable for negative local impacts on the environment and landscape, rural residents may perceive the burden of hosting them that urban residents do not have to shoulder. Nilson and Stedman [18] are among the few scholars who examined the role of this perceived rural burden on RE acceptance. They found that greater perceived distributive injustice between rural and urban areas (e.g., upstate and downstate New York) leads to a higher level of opposition to RE development. Moreover, in a separate study, they also find that rural opposition in upstate New York is primarily geared toward utility-scale solar energy facilities, quoting the residents’ fear of scenic, agricultural lands being replaced by “a commercial, industrial power plant” [63].

Rural exploitation also speaks to numerous studies that suggest the variants of energy injustice (i.e., procedural, distributive, restorative, and recognition) are key drivers of local opposition to RE development [44,64–68]. Rural poverty contributes to the power imbalance between rural residents and RE developers, often with abundant financial and political resources if they bring utility-scale RE facilities. This power imbalance makes it difficult for rural residents to participate meaningfully in the planning process. Therefore, they often cannot obtain knowledge about anticipated negative environmental and health-related impacts or agree on terms that channel benefits into their communities [69]. The lack of education and professional opportunities in rural communities [70,71] may further exacerbate the policy literacy that rural residents need to understand the project proposal and approval process. As a consequence, rural communities face a lack of effective means to communicate their diverse needs and perspectives about RE development, suffering from recognition bias (i.e., distorted or discriminative understanding of rural needs) or restorative injustice (i.e., failure to address a broader context of rural-urban inequality in both economic and RE development) [60].

2.2. Narratives of rural exploitation in renewable energy development

We define narratives as simplified stories of a complex problem that (often) suggest simplified solutions or prescriptions to a current situation that is deemed undesirable by the narrators. As Stone [72] put it,

narratives are about “how we [narrators] know there is a disparity between social goals and the current situation.” Narratives can play an important role in shaping local acceptance of RE facilities because they help individuals make quick assessments about their local impacts that are often long-term and complex. Scovell et al. [73] examined the narratives about the positive societal impact of RE transition and found that individuals who agree with such narratives show a higher level of acceptance for solar energy facilities. Because of the power of narratives, opponents of RE development may actively resort to their use to spur further mobilization. Fergen et al. [74] analyzed 803 posts uploaded on social network services (SNS) by two opposition groups based in northern Ohio, designed to disseminate (mis)information about the negative impacts of wind turbines on human health and public safety.

Narratives that undermine local RE acceptance may also convey a message that RE facilities exacerbate and entrench what is already an undesirable situation. This is why, for RE opponents in rural areas, narratives of rural exploitation by RE facilities might be an effective tool to mobilize further opposition from rural residents. These narratives would highlight that rural areas, especially underprivileged areas where residents may feel they have already been losing out, are nevertheless shouldering more burdens by hosting RE facilities and bearing their local negative impacts, with their benefits primarily flowing to urban areas. Moreover, these narratives may (often falsely) portray that urban ‘outsiders’ mainly propose and operate RE facilities in rural areas, the notion that Ki and Yun [75] showed to have undermined local RE acceptance.

In short, the narratives of rural exploitation will effectively decrease rural support for RE development in rural (and particularly underprivileged) areas. Such narratives might not always reflect the actual distribution of costs and benefits from RE development between rural and urban areas. Still, they might prove effective in mobilizing RE opposition in such rural areas for three reasons. First, they can nurture a sense of unfairness and injustice from RE development in rural residents that various studies have pointed out as key psychological factors undermining RE acceptance [76–78]. Second, a broader context of rural-urban inequality provides a historical and normative justification for rural residents to perceive unfairness and injustice from RE development, thereby strengthening their beliefs. Lastly, these narratives may motivate rural residents to distinguish themselves from urban populations or RE developers who (they perceive) sit at the opposite end of rural exploitation, thereby reinforcing the in-group identity and cohesion among rural opponents [79].

Yet, few studies have empirically tested the effectiveness of rural exploitation narratives in affecting rural support for RE development. One notable study by Aklin et al. [80] conducted a framing survey experiment in rural India where they included a treatment frame highlighting the unfairness of advantages of solar energy between rural and urban areas, but they did not examine whether the framing effect should vary across rural areas. Therefore, this study examines whether rural exploitation narratives reduce RE support among rural residents through field survey experiments in South Korea in two rural areas with different levels of economic development and remoteness.

2.3. History of rural exploitation in South Korea

South Korea is a promising area of research to address the research question above because it manifests a profound history of rural exploitation. Historically, South Korea’s economic growth initially relied on an export-led industrial policy, primarily on the labor-intensive manufacturing industry. Moore [34] argued that the success of this policy “was predicated upon the ‘squeezing’ of the agricultural sector” through keeping grain prices lower, which had led to the impoverishment of the rural agricultural sector. When the South Korean economy transitioned from labor-intensive to capital-intensive, heavy manufacturing industries, government investment was still mainly concentrated in the Seoul and Busan Metropolitan Area in the Southeast.

Rural impoverishment further persisted after the 1990s when the South Korean government opened the domestic agricultural market, which negatively impacted farmers with less global comparative advantage due to their small-sized farmlands [81,82].

Like in many places, rural exploitation in South Korea is not only confined to economic domains. Park and Sovacool [83] analyzed the siting of nuclear power plants and waste facilities in Ulju and Gyeongju, suggesting that contested energy facilities migrate to geographically remote communities that “lack the political, social, and economic strength to oppose them,” further ‘peripheralizing’ these communities in spatially uneven patterns of energy development. Rural areas in South Korea are also comparably burdened by RE development if one looks at the local RE self-sufficiency rate, calculated by how much electricity each area produces from RE compared to how much it consumes. Seoul Metropolitan City produced 0.66 % of its total electricity consumption from 6 RE sources (solar, wind, hydro, tidal, bioenergy, and waste), while rural provinces such as Gwangwon produced 30 %, and Jeonnam produced 15 % in 2020 [84]. This pattern aligns with the overall electricity production-consumption gap between rural and urban areas in South Korea, with the highest level of inequality in terms of the spatial unevenness of power generation among other countries studied by Perez-Sindin et al. [27].

These pieces of evidence suggest that South Korea has followed a trajectory of rural exploitation in economic and energy terms at the expense of rapid economic growth through export-led industrial policy. We now explain the background of two rural areas of study and elaborate on our central hypothesis.

3. Study background

All indicators to follow are based on the Korea Statistical Information Service (KOSIS) [85], a government-operated statistical database with up-to-date data on local-level sociodemographic and economic indicators. This study relies on the Electric Power Statistics Information System (EPSIS) [86] for energy-related indicators.

Wando in Jeonnam Province and Yeosu in Gyeonggi Province are rural counties in South Korea with a population density lower than the national average (520 per km²) as of 2020. Wando is home to 48,329 residents with 122 per km² of population density, and Yeosu is home to 115,358 with 190 per km². According to KOSIS, the urban area comprises residential, commercial, industrial, and recreational park areas. The urban area in Wando is 26.63 km² (6.7 % of the total area), and that in Yeosu is 26.49 km² (4.3 %).

Both counties have hosted a comparable amount of RE facilities since 2012 and hence demonstrated rural opposition to RE development, as shown in restrictive land-use ordinances against solar energy facilities adopted by their local governments. In Wando, the total RE capacity installed and in operation in 2020 was 101.72 MW (with a solar capacity of 84.19 MW; 0.17 kW per capita), producing 152,603 MWh of electricity (about 14 % RE self-sufficiency rate). In Yeosu, the total RE capacity in 2020 was 134.69 MW (121.79 MW of solar; 0.10 kW per capita), producing 208,684 MWh of electricity (about 15 % RE self-sufficiency rate). Yeosu ranks itself as the second county in Gyeonggi Province in terms of solar energy capacity, while Wando is below average in Jeonnam Province as it is populated with counties with solar energy capacity far higher than the national average. This comprises a primary reason for case selection: Wando and Yeosu, despite many differences to follow, are comparable in terms of population density and urban area percentage, and most importantly, solar energy capacity (particularly in terms of per capita) and RE self-sufficiency rate.

Key differences between the two areas of study are geographical remoteness and economic indicators. As visualized in Fig. 1, Wando, an island county located in the Southwestern region of South Korea, is distanced away from Seoul Metropolitan City by 362 km. The closest metropolitan city, Gwangju, is 95 km away from Wando, but due to rigged coastal and island terrains, it takes about 2 h to drive and 3 h of

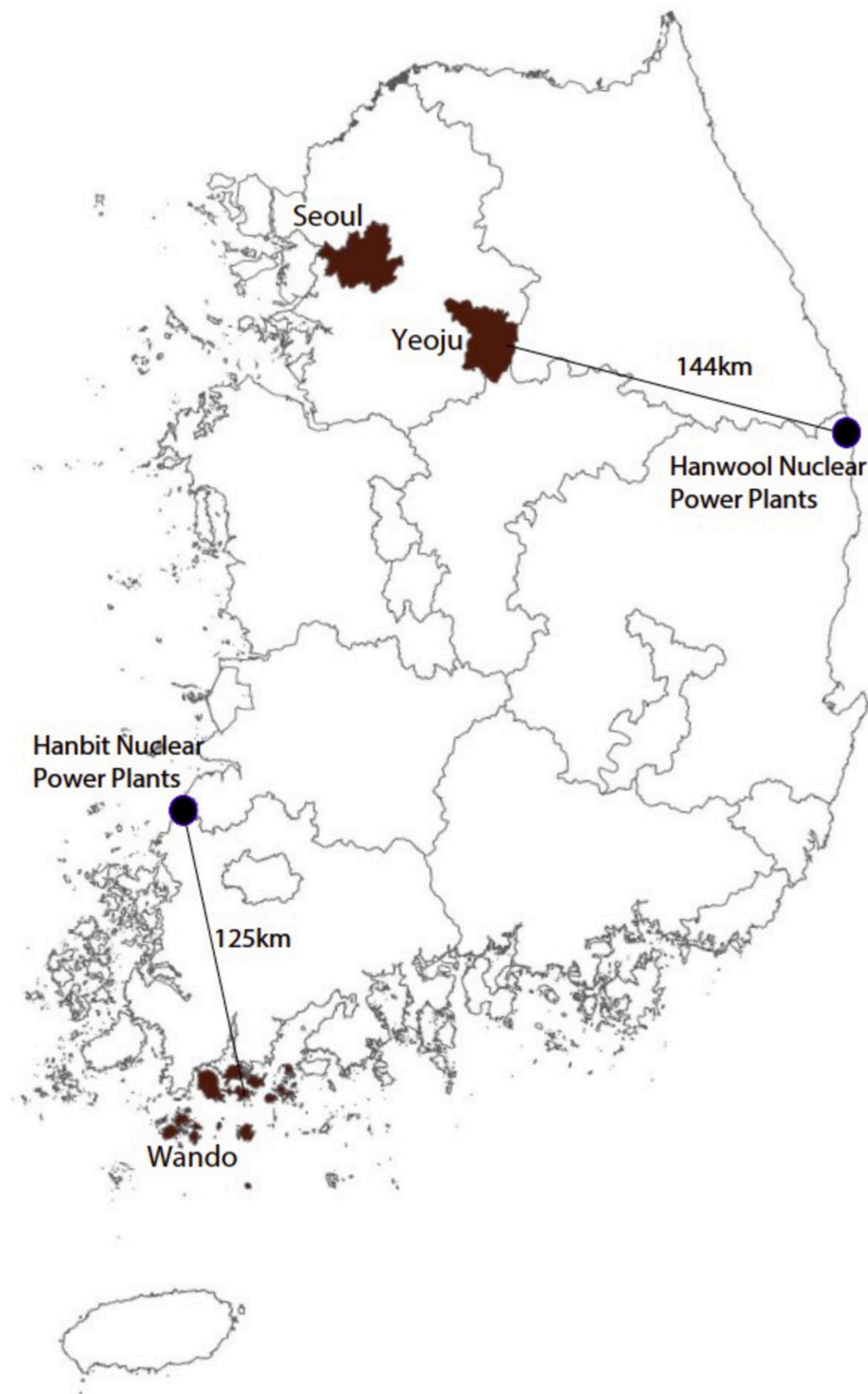


Fig. 1. Survey areas.

public transportation between the two areas. Yeosu, however, is located 62 km away from and connected to Seoul Metropolitan City through the subway (Line 1). While both Wando and Yeosu are comparably burdened with RE facilities, particularly by solar energy facilities, this geographical difference suggests that these two areas exhibit a varying level of ruralness.

Another difference lies in economic indicators. Wando's gross regional domestic production (GRDP) as of 2021 is 1183 trillion Korean Won (859.7 million USD), and Yeosu's GRDP is 4663 trillion Korean won (3388.8 million USD). GRDP per capita of Wando is thus 17,789 USD,

whereas that of Yeosu is 29,376 USD. This difference in part is attributable to the configuration of economic sectors in these two areas: Wando's economy is primarily dependent on the public (24 %) and agricultural sectors (19 %), whereas the first and second largest economic sectors of Yeosu are manufacturing (21 %) and real estate businesses (9 %). This implies Wando is a more economically underprivileged rural county than Yeosu. [Table 1](#) summarizes the comparison between these two areas.

Given the background information of the study areas, we reintroduce our research question: Would the narratives highlighting the rural

Table 1
Comparison of Wando and Yeosu.

	Wando	Yeosu
Population size	48,329	115,358
Population density	112.00	189.80
Solar capacity (MW)	84.19	121.79
Solar capacity per capita (kW)	0.17	0.10
Electricity consumption (MWh)	1,089,338	1,407,366
RE electricity production (MWh)	152,603	208,684
RE self-sufficiency rate (%)	14.00	14.83
Solar electricity production (MWh)	113,865	137,249
Solar self-sufficiency rate (%)	10.45	9.75
GRDP (million USD)	859.70	3,388.82
GRDP per capita (USD)	17,789	29,376

exploitation of RE development affect RE support among rural residents? More specifically, would they render different effects on residents' support depending on how underprivileged their rural area is? This leads to our main hypothesis: *Wando residents will show lower support toward RE development when they are exposed to the narratives priming rural exploitation than Yeosu residents.* We now turn to the empirical strategy to test this hypothesis.

4. Data and methods

4.1. Survey design

This study draws on a field survey experiment with vignettes randomly assigned to survey participants in Wando and Yeosu (IRB No: STUDY00015587). The survey took place between September and November 2022. Ideally, the experiment must cover all villages of these two counties to ensure representativeness. There were practical challenges to this approach. First, the survey was conducted in person with a limited budget and timeline. An in-person survey was particularly crucial for the context of this study because the low level of digital literacy in rural areas might challenge survey participants. Also, residents in the survey areas were cautious about exposure to COVID-19, especially when they were more elderly, which made it difficult for researchers to recruit a wide range of survey participants across many areas. Therefore, the survey focused on one village in each county, namely Yaksan in Wando and Yeoheung in Yeosu. The researchers, however, have tried to cover as many sub-localities (townships) within each village as possible. Survey participants were recruited by canvassing and visiting community centers.

Yaksan is a village in Wando with 21 townships dispersed across the main island, Joyak. The village is known for a 180-MW utility-scale solar energy project proposed by a South Korean multinational corporation, SK E&S, but not yet fully approved. If approved, the project is expected to be the largest solar energy facility in South Korea on 400-acre land. The corporation plans to build the facility on a large rice cropland landfilled in 1997 to boost rice production in the area, which was difficult due to the mountainous terrain of the main island. The cropland was accused of lower agricultural productivity due to its salinity level, which provided room for the corporation to propose the project replacing the cropland. This proposal has spurred a divisive contestation among residents, splitting them between supporters and opponents of the project.

Yeojeung is a village in Yeosu with 12 sub-localities in the southeast part of Yeosu. As an administrative and business center of Yeosu, the area is home to public gatherings attracting residents from other villages of Yeosu. While Yeosu has had no proposals for a comparable size of utility-scale solar energy project as that in Yaksan, there have been a noticeable number of cases where residents opposed solar energy projects in their neighborhood. For instance, Jeomdong village, adjacent to Yeoheung, became a contested area where residents protested against a solar energy project proposed in one of its townships in 2019. It is noteworthy to

reiterate that both counties have implemented a land-use restriction on solar energy facilities since 2018, placing 500-m setback requirements on them from nearby roads and households in response to local petitions citing deforestation and landscape disruption [24].

Since the primary type of RE facilities in both areas is solar, this study focuses on solar energy. The survey randomly assigned one of the three vignettes that convey information on solar energy facilities, as shown in Table 2. The first two vignettes serve as treatments designed to prime rural exploitation of solar energy development. Specifically, the first vignette highlights that solar energy development in rural areas serves the urban needs for the consumption of electricity from solar energy facilities. The second vignette highlights that urban RE developers enjoy most of the profits from solar energy development. These vignettes address two possible types of rural exploitation of RE development that rural residents might perceive: one by urban electricity consumers in general and the other by RE developers more specifically. Survey participants exposed to the third vignette are in a reference group as it does not contain any narrative about rural exploitation. The survey also included three attention-check questions. All vignettes are designed to have the same word counts in Korean.

A field survey experiment can be susceptible to violating the stable unit treatment value assumption (SUTVA). In our context, if survey participants who have completed the survey interact with potential participants who may be invited to the study later, they may share information about their vignettes and questions, causing unmodelled spillovers. To prevent this, researchers recruited participants from one township for one day at a time and did not recruit any further from the township already visited. This is because each township can be 'contaminated' after the survey. For instance, after the survey garnered responses from 15 respondents in Woodoo-ri (township) in Yaksan in one day, the survey no longer took place there. The next round of the survey took place in, say, Haedong-ri, where residents do not frequently interact in person with residents in Woodoo-ri because of the distance between these two villages. Even when gathering responses in one township, researchers asked the participants not to reveal any information about the survey to anyone.

After reading the vignette, survey participants were asked to respond to the following question, which serves as the main outcome of interest:

"To mitigate greenhouse gas emissions, the South Korean government plans to invest in (1) expanding new solar energy facilities and (2) renewing existing nuclear energy facilities. If it were to invest 10

Table 2
Survey vignettes and their narratives.

Vignette	Narrative
Vignette 1 (Rural exploitation by urban consumers)	Solar energy facilities help mitigate climate change by producing less amount of greenhouse gas emissions than fossil fuel energy to generate the same amount of electricity. Meanwhile, solar energy facilities are largely located in rural areas, but they send most of their electricity to urban areas for their residents to consume.
Vignette 2 (Rural exploitation by urban RE developers)	Solar energy facilities help mitigate climate change by producing less amount of greenhouse gas emissions than fossil fuel energy to generate the same amount of electricity. Meanwhile, solar energy facilities are largely located in rural areas, but urban developers earn most of the profits from selling their electricity to the electricity market.
Vignette 3 (Reference)	Solar energy facilities help mitigate climate change by producing less amount of greenhouse gas emissions than fossil fuel energy to generate the same amount of electricity. Meanwhile, there are two types of solar energy facilities: those that use photovoltaic (PV) panels, and those that use concentrated heat from the Sun.

billion Korean Won, in which option do you think more investment should be made and by how much?"

This survey question asked participants which low-carbon option, between solar energy investment and nuclear energy renewal, they favored more than the other. This approach draws on Aklin et al. [69], who also indicated a trade-off between two government investment options in their survey questionnaire. They revealed that their pilot study without a trade-off received an overwhelmingly positive response toward solar energy subsidies because respondents might prefer solar energy investment if it entails no opportunity costs from choosing an alternative energy investment. In our survey context, South Koreans have widely conceived the trade-off relationship between solar and nuclear energy because previous administrations, namely the Lee Myung-Bak and the Moon Jae-In administrations, had completely different policy approaches toward these sources. While the Lee Myung-Bak (and Park Geun-Hye to a lesser extent) administration adopted a pro-nuclear energy policy, the Moon Jae-In administration promoted an ambitious RE policy in pursuit of a 'nuclear phase-out' [87]. The scholarship also widely discusses the trade-off relationship between these two sources [88–90]. Therefore, our survey also imposes a trade-off in the questionnaire that measures the level of support for solar energy to prevent overtly positive responses and reflect the political context of solar-nuclear energy policies in South Korea.

This approach, however, has several caveats. Even though the vignettes do not contain any information on nuclear energy, survey participants might respond differently to this question based on their prior preference for nuclear energy. Hence, this study addresses this concern in the following steps. First, we selected both counties and sub-localities where the survey took place to have a comparable distance from the closest nuclear energy power plant. This ensures that the proximity to the nuclear power plant does not explain the difference in response patterns between the two areas of study. Specifically, Yeoheung is located 144 km from the closest nuclear power plant, Hanwool Nuclear Power Plant, and Yaksan is located 125 km away from the closest one, Hanbit Nuclear Power Plant, with both distances costing about the same time to drive. Second, because the pro-nuclear administration was politically right-wing and the pro-solar administration was left-wing, the survey asked the self-reported political ideology of participants on a 1–5 scale (1-very liberal; 5-very conservative) after the vignettes and the question above. This is particularly important in the South Korean context, where conservative political ideology is highly correlated with pro-nuclear attitudes [91]. Moreover, the survey also asked how much participants agreed with the following statement on a 1–5 scale (1-strongly disagree; 5-strongly agree): "How much do you agree nuclear energy is a more reliable source of electricity over renewable energy?" (hereafter 'Nuclear Reliability Question') This question aimed to examine participants' predisposed beliefs on the trade-off relationship imposed in the main question above. The self-reported political ideology and the level of agreement with this statement are used as control variables and inspected in the balance statistics to ensure they do not influence the estimated treatment effects of the vignettes.

The remaining section of the survey asked for participants' prior beliefs about rural exploitation in general and their sociodemographic background, which might explain their prior preferences toward solar energy. First, the survey asked again how much they agreed with the following statements: "How much do you agree the government policy has been in favor of urban areas as opposed to rural areas?" (hereafter 'Urban Bias Question') and "How much do you agree there is a widening gap between urban and rural areas in terms of economic growth?" (hereafter 'Rural-Urban Gap Question') The first question directly surveys the participants' perception of the 'urban bias' in government policy [92], and the second question asks about their belief in the relative poverty of rural areas compared to urban areas. In line with our theoretical discussion, even within the same area of study, rural residents who more strongly agree with these statements would be more

likely to show less support toward solar energy investment if exposed to the first or the second vignettes than those who less strongly agree.

Next, one sociodemographic information theoretically important for our study is the level of educational attainment. Although there is no available data on educational attainment at a local level in South Korea, the survey asked this information to examine whether lower policy literacy associated with the lack of educational opportunities motivates rural residents to sympathize more strongly with the narratives of rural exploitation of RE development. Other sociodemographic information the survey asked for includes age, gender (as shown in their government ID), and monthly household income, which is also theoretically relevant to the effect of rural exploitation narratives.

4.2. Empirical analysis

This study used an ordinary least squares (OLS) model to examine the effects of rural exploitation narratives on solar energy support, which experimental studies widely use when the outcome variable is continuous. The average treatment effect (ATE) is captured in the mean estimate in the participants' response to the main outcome question, e.g., 7 billion on solar energy investment (which leaves 3 billion on the other option) between two treated groups and one control group (e.g., Vignette 1 vs Vignette 3 groups, Vignette 2 vs Vignette 3 groups).

To check any potential biases in the ATE estimates, this study conducted a balance check across survey groups regarding responses to survey questions and sociodemographic background using a one-way ANOVA test. Each variable used in the analysis goes through this test to check if survey groups are statistically distinguishable regarding the variable. Specifically, if the *p*-value is lower than 0.05, we reject the null hypothesis that the three means of each variable across groups are statistically different. This means that the random assignment of the vignettes cannot be validated.

To reiterate our main hypothesis, we expect that Yaksan residents will show lower support toward RE development when they are exposed to the vignettes priming rural exploitation than Yeoheung residents. Hence, we devised three models: Model 1 uses a full sample of all survey participants in two counties, Model 2 uses a subsample of Yaksan residents, and Model 3 uses a subsample of Yeoheung residents. We expect that Model 2 renders a stronger negative ATE estimate than that in Model 3.

Finally, we acknowledge another source of threat to the validity of our ATE estimates: post-treatment bias [93]. Specifically, the 'Urban Bias Question' and 'Rural-Urban Gap Question' are susceptible to this bias because the first and the second vignettes are designed to prime residents with rural exploitation in solar energy development. Hence, residents exposed to these vignettes might respond differently to these questions from those exposed to the third vignette (reference group). Therefore, we use the sensitivity analysis [94] to see whether the original results hold if the effects of responses to these two questions are assumed to be stronger than what appears on the baseline results.

5. Results

5.1. Descriptive statistics

The survey invited 389 survey participants in total, returning an acceptance rate of 61.7 % (240 survey responses collected). Among these responses, 55 of them were returned as incomplete as participants ceased to participate in the survey, which obtained the informed consent of voluntary participation. Among 195 complete responses, 34 of them failed to pass the attention check questions, leaving 151 effective samples for analysis, as outlined in Table 3.

The number of effective responses is crucial for the OLS estimation because standard errors of the coefficient estimates rely on the size of the sample. In our case, all experimental groups in two study areas have a sample size smaller than 30, which might induce serious inefficiencies (i.

Table 3

The number of responses in each county.

	Yaksan, Wando			Yeoheung, Yeoju			Total
	Vignette 1	Vignette 2	Vignette 3	Vignette 1	Vignette 2	Vignette 3	
Total responses	36	45	39	44	34	42	240
Incomplete responses	6	13	9	10	9	8	55
Attention check failures	4	6	3	8	6	7	34
Effective responses	26	26	27	26	19	27	151

e., high mean squared errors) in the ATE estimates. Therefore, along with the OLS results, this section presents results using a Bayesian approach that relies less on the sample size [95] but with the same model specification.

Table 4 summarizes descriptive statistics and *p*-values from a one-way ANOVA test for balance check purposes. The balance statistics suggest that there is no significant baseline difference in all variables across treatment and control groups. Comparing all six groups across two survey areas renders the same result. This suggests that the experiment ensured the random assignment of vignettes to survey participants. Moreover, the vignette a respondent is exposed to does not predict whether the respondent has failed the attention check. The one-way ANOVA test between whether a respondent failed it and what vignette respondents received fails to reject the null hypothesis (*p*-value = 0.776).

5.2. Main results

Table 5 summarizes the results from three OLS models. Using the full sample, Model 1 results suggest that both the first and the second vignettes negatively affected respondents' solar energy support. When examining Model 2, however, we find Yaksan residents primarily drive these results. Survey participants in Yaksan showed less support when exposed to either the first or second vignette. Specifically, Yaksan residents exposed to the first or the second vignette priming rural exploitation allocated about 1.8 billion Korean Won of government investment less to solar energy than those exposed to the reference vignette. In Yeoheung, however, survey participants were indifferent to what vignette they received. On the other hand, the result in Model 1 regarding the 'Nuclear Reliance Question' seems to be primarily driven by Yeoheung residents. Survey participants in Yeoheung, who more strongly agree with the advantage of nuclear energy in reliability over solar energy, allocated 0.9 billion less to solar energy than those exposed. In light of potential post-treatment bias, we analyzed the data

Table 5
OLS results.

Variables	Model 1	Model 2	Model 3
Sample	Full	Yaksan	Yeoheung
Vignette 1: Rural exploitation by urban electricity consumers	−11.98** (5.38)	−18.73*** (6.87)	0.59 (8.08)
Vignette 2: Rural exploitation by urban solar energy developers	−8.03** (5.41)	−18.04*** (7.29)	0.94 (8.02)
Nuclear Reliance Question	−5.31*** (2.14)	−1.02 (3.19)	−9.23*** (2.55)
Urban Bias Question	4.28* (2.73)	5.07 (3.19)	4.39 (4.50)
Rural-Urban Gap Question	1.58 (2.96)	−1.32 (3.59)	2.63 (5.13)
Political Ideology	−3.71 (2.64)	−4.59 (2.89)	−4.67 (4.60)
Education	3.77** (2.20)	2.95 (3.30)	−1.26 (3.08)
Age	−0.08 (0.17)	−0.19 (0.28)	−0.23 (0.27)
Gender	−0.12 (3.64)	−3.33 (4.94)	−5.52 (4.79)
Household Income	0.36 (1.61)	−0.17 (2.10)	−0.41 (4.79)
Intercept	42.42*** (17.77)	51.80*** (25.19)	92.88*** (0.00)
Adjusted R-squared	0.1755	0.1628	0.2168
N	151	79	72

Note: Heteroskedasticity-adjusted standard errors (HC4) are reported in parentheses.

*** *p* < 0.01.

** *p* < 0.05.

* *p* < 0.1.

without two survey questions, as explained above, and the results remain the same. This lends support to our main hypothesis.

Next, we re-analyzed Models 1–3 with a Bayesian approach, given that the small number of observations in each treatment-area group

Table 4

Descriptive statistics with balance checks.

Variables	Yaksan				Yeoheung			
	Vig 1	Vig 2	Vig 3	P-val	Vig 1	Vig 2	Vig 3	P-val
Nuclear Reliability Question	3.27 (1.00)	2.69 (1.19)	3.04 (1.13)	0.176	3.12 (1.24)	3.21 (1.47)	3.22 (1.22)	0.950
Urban Bias Question	4.15 (0.78)	3.81 (1.41)	3.78 (1.31)	0.457	3.85 (1.05)	3.95 (1.18)	3.96 (1.10)	0.919
Rural-Urban Gap Question	4.04 (0.96)	3.62 (1.24)	3.78 (1.19)	0.403	3.88 (1.18)	4.32 (0.58)	4.19 (0.83)	0.270
Political Ideology	2.69 (0.62)	2.85 (1.05)	2.74 (0.94)	0.816	2.96 (0.96)	2.74 (0.73)	2.93 (0.78)	0.648
Education	4.19 (1.47)	3.65 (1.35)	3.67 (1.71)	0.349	5.08 (1.44)	5.05 (1.27)	4.59 (1.25)	0.345
Age	53.5 (14.8)	60.4 (16.0)	53.8 (15.4)	0.195	49.8 (15.3)	55.9 (13.0)	56.1 (12.0)	0.186
Gender	0.73 (0.45)	0.54 (0.51)	0.48 (0.51)	0.162	0.69 (0.47)	0.79 (0.42)	0.74 (0.65)	0.834
Household Income	2.27 (1.04)	2.23 (1.31)	2.56 (1.58)	0.624	2.85 (1.62)	3.11 (1.37)	3.59 (1.80)	0.248

Note: Cell entries are subgroup means for each variable with standard deviation reported in parentheses. P-value (P-val) is derived from the one-way ANOVA test. 'Vig' stands for 'vignette' (e.g., Vig 1 = Vignette 1 group).

might have provided inefficient estimators in the original results [96]. We employed the Hamiltonian Monte Carlo approach [97] to obtain the posterior distribution of the parameters (i.e., coefficient estimates), running four separate Markov chains with 5000 iterations that provide 20,000 posterior draws for each parameter. All chains indicate convergence using the Gelman-Rubin convergence diagnostic [98]. We used an informative prior for parameters using coefficient estimates and standard errors from the baseline OLS results. The original results hold. Table 6 summarizes these results with median and 95 % credible intervals of the posterior distributions.

The use of the Bayesian approach is far from a panacea to small sample size problems, however, given that the Bayesian approach is highly sensitive to the specification of the prior distribution [99]. In this context, the informative prior explained above might produce results biased toward the baseline OLS results. Therefore, we used an uninformative prior, and the results remain consistent.

5.3. Sensitivity analysis

Finally, we interrogate the findings of Model 2 with sensitivity analysis. The analysis shows whether the treatment effects in Model 2 hold if we assume the size of the confounding effect is stronger (typically

by two to three times). Specifically, we assumed the effects of the ‘Urban Bias Question’ and ‘Rural-Urban Gap Question’ are stronger than what they are in the baseline model and examined whether the treatment effects (of the first and the second vignettes) are robust to these assumptions.

Table 7 shows how the treatment effects in Model 2 change under each assumption. We show that all treatment effects adjusted under each sensitivity assumption are consistent with those in the baseline model. This implies that while Yaksan residents could have varying responses to the ‘Urban Bias Question’ and ‘Rural-Urban Gap Question,’ these responses did not significantly confound the relationship between treatments and the outcome in our model.

6. Discussion

As pointed out earlier, few studies have explored what narratives can undermine local acceptance of RE development, particularly in rural areas where a number of cases of RE opposition have been reported. This study provides evidence that the narratives that prime rural exploitation can undermine rural acceptance of RE development. This finding speaks to the literature that investigates various sources of rural opposition to RE development, particularly from an energy justice perspective.

Table 6
Bayesian linear regression results.

Variables	Model 1a	Model 2a	Model 3a
Vignette 1: Rural exploitation by urban electricity consumers	−12.0 [−19.2, −4.9]	−18.7 [−27.9, −9.5]	0.6 [−9.8, 10.7]
Vignette 2: Rural exploitation by urban solar energy developers	−8.0 [−15.0, −0.9]	−18.0 [−27.3, −8.7]	1.0 [−8.8, 10.5]
Nuclear Reliance Question	−5.3 [−8.1, −2.6]	−1.0 [−5.2, 3.2]	−9.2 [−12.5, −5.8]
Urban Bias Question	4.3 [1.0, 7.7]	5.1 [1.0, 9.1]	4.4 [−0.7, 9.4]
Rural-Urban Gap Question	1.6 [−2.2, 5.2]	−1.3 [−5.9, 3.2]	2.7 [−3.2, 8.3]
Political Ideology	−3.7 [−7.5, −0.1]	−4.6 [−8.7, −0.4]	−4.7 [−10.9, 1.4]
Education	3.8 [1.3, 6.2]	2.9 [−0.6, 6.5]	−1.3 [−5.1, 2.6]
Age	−0.1 [−0.3, 0.1]	−0.2 [−0.5, 0.1]	−0.2 [−0.6, 0.1]
Gender	−0.1 [−5.5, 5.3]	−3.4 [−10.5, 3.7]	−5.6 [−12.4, 1.4]
Household Income	0.4 [−1.7, 2.4]	−0.2 [−3.3, 3.0]	−0.4 [−3.1, 2.3]
Intercept	42.5 [17.5, 67.0]	52.0 [18.0, 86.2]	93.5 [54.7, 134.4]
N	151	79	72

Table 7
Sensitivity analysis results.

Treatment	Sensitivity assumption	Adjusted effect
Vignette 1: Rural exploitation by urban electricity consumers	Urban Bias Question, 2×	−17.99 [−31.53, −4.44]
	Urban Bias Question, 3×	−17.61 [−30.91, −4.31]
	Rural-Urban Gap Question, 2×	−18.63 [−32.69, −4.66]
	Rural-Urban Gap Question, 3×	−18.58 [−32.57, −4.59]
Vignette 2: Rural exploitation by urban solar energy developers	Urban Bias Question, 2×	−15.17 [−28.76, −1.59]
	Urban Bias Question, 3×	−13.70 [−27.14, −0.27]
	Rural-Urban Gap Question, 2×	−17.80 [−31.65, −3.94]
	Rural-Urban Gap Question, 3×	−17.68 [−31.53, −3.82]

Studies in this literature highlight that perceptions of distributive injustice drive local opposition to RE development [66,100], and, especially in a rural setting, this distributive injustice means the unfair distribution of costs and benefits from RE development between rural and urban areas [18]. Our finding contributes to this literature by suggesting that narratives of rural exploitation can nurture and reinforce this perception of distributive injustice, decreasing rural support for RE development.

While it is difficult to generalize the results from our survey data to county-level ones, the authors note that Yaksan and Yeohung are fairly representative cases of villages in Wando and Yeosu, respectively, as sub-localities in rural counties in South Korea are not economically heterogeneous as those in urbanized counties are. With this being addressed, we highlight that the effect of the narratives was mainly driven by survey participants in Wando, which various indicators showed to be more economically underprivileged than Yeosu. This is because residents in such underprivileged rural areas may perceive a greater economic inequality between rural and urban areas and, therefore, may resonate with the notion of rural exploitation more strongly. This implies that rural residents do not always need to directly experience the negative externalities of RE facilities to become their opponents. Even without such experiences, the narratives that convey the idea of rural exploitation can negatively change the view of rural residents toward RE development. As previous studies put it [4,53], our finding confirms that rural opposition can be more than just a NIMBY-response. These narratives may mobilize opposition in rural areas with and without RE projects, particularly if such areas are economically underprivileged. Even though this pattern is subject to future empirical scrutiny, numerous cases and studies show that rural RE opponents (or groups) actively utilize such narratives.

Our finding also speaks to the body of literature that explains why RE developers' efforts to channel the benefits from RE projects into local communities, such as compensation schemes, do not often work to enhance their local acceptance [101–104]. While it is fair to point out that some compensation schemes fail to consider indivisible, non-monetary values that local residents cherish [102], the notion of rural exploitation might further help explain why sometimes any compensation schemes offered by RE developers are deemed unfair by rural residents. This is because rural residents can view that any RE projects and their compensation schemes, if framed as practices of rural exploitation, would simply deepen rural-urban inequalities. In other words, the strength of perceived rural exploitation, rather than the design of compensation schemes, might better explain the level of local RE acceptance.

The narratives of rural exploitation would make it even more difficult for RE developers to work with local RE acceptance because these narratives motivate rural opponents to view RE developers as outsiders, which reinforces their low level of trust toward RE developers [75]. In such cases, overcoming the image of outsiders can be an initial step for RE developers to work with local RE acceptance, such as through extensive research on the historical and sociocultural background of opponents or their communities. Yet, a recent study by Nilson et al. [29] shows while RE developers are increasingly aware of the importance of community acceptance (or opposition), they are less willing to engage in continued interactions with the community that would help understand their needs and motivations. However, this may be largely because there is a lack of research on the effective models of community engagement for local RE acceptance, particularly that highlights the historical understanding of local communities. Future research should investigate what models of community engagement best help RE developers overcome the image of outsiders, especially when the notion of rural exploitation characterizes the local opposition they face.

7. Conclusion

This study employed a field survey experiment in two study areas in

South Korea to examine whether the narratives of rural exploitation affect rural support for solar energy development. Our results show that such narratives decrease support, but only among residents in Yaksan, Wando, which is more economically underserved than Yeohung, Yeosu. Based on our theoretical discussion, we suggest that this is because rural residents in more economically underserved areas may perceive a greater rural-urban inequality and, therefore, more strongly resonate with the notion of rural exploitation. Our results are robust to additional analyses such as a Bayesian approach and sensitivity analysis.

It is out of the scope of this study to argue that RE development, particularly solar, is always retrenching the pattern of rural exploitation by urban or it is a primary driver of all cases of rural opposition to RE development. As outlined in the introduction, the perception of rural exploitation may not represent all rural opponents' and non-opposing rural residents' views toward RE development. However, we contend that the narratives of rural exploitation can be an effective tool for mobilizing opposition among residents in rural areas that lack access to economic and political resources. If such a lack of access is a product of a broader context of rural-urban inequality or historical patterns of rural exploitation, the effect of such narratives might be stronger.

This is why, in many cases of rural opposition to RE development, such narratives can be popular among opponents. The notion that RE development in rural areas is a continuum of long-standing rural exploitation can promote a sense of unfairness and injustice that can decrease RE acceptance. This notion could also serve as an effective counterargument to the pro-RE narrative of tackling climate change and enhancing domestic energy self-reliance [105], the narrative of which critics often conceive rural opposition as a NIMBY-ism [3,4]. Viewed in this way, the narratives of rural exploitation are tools that rural opponents may use to overcome recognition bias in RE development toward them who, though to a varying degree across countries, face a growing burden of negative externalities from RE facilities primarily concentrated in rural areas [27]. Also, as rural areas (will) host more RE facilities, rural opponents will find it more attractive to use the narratives of rural exploitation to reflect a widening mismatch in RE development between rural and urban areas.

We conclude by pointing to the areas of future research. First, even if this study conducted an individual-level survey experiment, we could not examine to which individual the narratives of rural exploitation would resonate more powerfully, and whether such individuals would more strongly oppose RE development. Therefore, individual-level effects of rural exploitation narratives on their RE support are open for future scrutiny. One could conduct a multi-site survey experiment with a mediation analysis to follow to investigate them further. Next, the number of respondents is one of the major limitations of this study that a Bayesian approach cannot fully overcome. Furthermore, while the main argument of the study relies on the socioeconomic differences between Wando and Yeosu, the data used to test it was mainly collected in one sub-locality in each county, leaving the generalizability of its results questionable. In light of this concern, researchers may benefit from previous works such as Webborn et al. [106], in which the authors elaborate ways to increase response rates in the context of energy research. Finally, based on a low explanatory power of our baseline OLS results, as indicated by the adjusted R-squared values, we acknowledge that there are multiple factors that explain rural support for RE development beyond the sense of distributive injustice. Building upon numerous studies that point out various sources of rural opposition to RE development, such as concerns over landscape disruption, future research may consider examining different weights across these sources in one study in terms of how strongly they can explain rural RE support.

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CRediT authorship contribution statement

Inhwan Ko: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

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Data availability

R codes and data of this study can be found online at: <https://doi.org/10.7910/DVN/YKLAEW>.

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